Banana Financial Intelligence Service

A production-grade RAG (Retrieval-Augmented Generation) system for grounded financial query answering with hallucination prevention, deployed on Kubernetes.

Prerequisites

Make sure the following are installed before you begin:

Tool	Version	Install
Docker	20+	https://docs.docker.com/get-docker
Minikube	1.32+	https://minikube.sigs.k8s.io/docs/start
kubectl	1.28+	https://kubernetes.io/docs/tasks/tools
Python	3.12	https://www.python.org/downloads

Step 1 — Get an Alpha Vantage API Key

The market data server requires a free API key from Alpha Vantage.

1. Go to https://www.alphavantage.co/support/#api-key
2. Fill in the short form — key is generated instantly
3. Export it in your terminal:

```bash
export ALPHA_VANTAGE_API_KEY=your_key_here
```

> Free tier: 25 requests/day, 5 requests/minute — enough for testing.

Step 2 — Start Minikube

```bash
minikube start --cpus=4 --memory=6144 --disk-size=20g
```

Verify it is running:

```bash
minikube status
```

Step 3 — Deploy to Kubernetes

Run the deploy script from the project root:

```bash
chmod +x deploy.sh
./deploy.sh
```

The script will:

- Point Docker at Minikube's internal daemon
- Build all 4 images: `banana-api:v2`, `banana-market:v1`, `banana-sec:v1`, `banana-social-mcp:v1`
- Create the Alpha Vantage secret in Kubernetes
- Apply all manifests (ConfigMap, MCP services, API deployment)
- Wait for all pods to become ready

> **First run:** The API pod downloads FinBERT and TinyLlama on startup (~1–2 GB). Allow 3–5 minutes.

Step 4 — Verify All Pods Are Running

```bash
kubectl get pods
```

Expected output:

```
NAME                    READY   STATUS    RESTARTS
banana-api-xxx          1/1     Running   0
banana-market-xxx       1/1     Running   0
banana-sec-xxx          1/1     Running   0
banana-social-xxx       1/1     Running   0
```

If any pod is not `Running`, check its logs:

```bash
```

```
kubectl logs deployment/banana-api
kubectl logs deployment/banana-market
kubectl logs deployment/banana-sec
kubectl logs deployment/banana-social
```

Step 5 — Access the API

On macOS with the Docker driver, use `port-forward` for a stable local URL.

Open a dedicated terminal and keep it running:

```bash
kubectl port-forward service/banana-api-service 8080:80
```

The API is now available at `http://localhost:8080`.

Step 6 — Test the API

Quick smoke test

Open a second terminal and run:

```bash
curl -s -X POST http://localhost:8080/analyze \
  -H "Content-Type: application/json" \
  -d '{"query": "What is AAPL stock price?"}' | python3 -m json.tool
```

Expected response:

```json
```

```
{
  "answer": "AAPL (2026-03-03) — Open: $263.32, High: $265.56, Close: $263.75 ...",
  "grounded": true,
  "hallucination_rate": 0.0,
  "faithfulness_score": 1.0,
  "block_reason": null,
  "tools_used": ["market", "sec"],
  "sentiment": {
    "label": "neutral",
    "confidence": 0.921
  },
  "report": "\nFinancial Report:\n\nSentiment: neutral\nConfidence: 0.921\n"
}
```


Full integration test suite

```bash
python3 test_service.py
```

Runs 17 queries across 6 categories and prints a summary report.

Benchmark (baseline vs optimized)

```bash
python3 benchmark.py --url http://localhost:8080

# Quick run with 4 queries only
python3 benchmark.py --url http://localhost:8080 --max 4

# Save results to CSV
python3 benchmark.py --url http://localhost:8080 --out results.csv
```

Rebuilding After Code Changes

If you update `main.py` or any agent file:

```bash
```

```
# Step 1 — Point Docker at Minikube's daemon
eval $(minikube docker-env)

# Step 2 — Rebuild the image
docker build -t banana-api:v2 .

# Step 3 — Restart the pod
kubectl rollout restart deployment/banana-api
kubectl rollout status deployment/banana-api
```

Confirm the new code is running:

```bash
kubectl logs deployment/banana-api | grep -E "Analyst|Scribe|Reflection"
```

Expected:

```
[API] Initialising AnalystAgent (loading FinBERT)…
[API] AnalystAgent / ReflectionAgent / ScribeAgent ready ✓
```

Useful Commands

Check pod and service status

```bash
kubectl get pods
kubectl get services
```

Stream live logs

```bash
kubectl logs -f deployment/banana-api
kubectl logs -f deployment/banana-market
kubectl logs -f deployment/banana-sec
kubectl logs -f deployment/banana-social
```

Filter logs by component

```bash
```

```
kubectl logs deployment/banana-api | grep "\[Fetch\]"
kubectl logs deployment/banana-api | grep "\[Analyst\]"
kubectl logs deployment/banana-api | grep "BLOCKED"
```

Restart a specific pod

```bash
kubectl rollout restart deployment/banana-api
```

Stop Minikube

```bash
minikube stop
```

Common Issues

Port 8080 already in use

```bash
lsof -i :8080
kill -9 <PID>
```

Pod stuck in Pending or CrashLoopBackOff

```bash
kubectl describe pod -l app=banana-api
kubectl logs deployment/banana-api --previous
```

New image not picked up after rebuild

Ensure you ran `eval $(minikube docker-env)` before `docker build`. Without this, Docker builds into your local daemon instead of Minikube's.

```bash
eval $(minikube docker-env)
docker images | grep banana-api   # check timestamp is recent
kubectl rollout restart deployment/banana-api
```

Market data returning empty results

Your Alpha Vantage key may not be set or the free tier daily limit (25 req/day) has been reached.

```bash
kubectl get secret alpha-vantage-secret -o jsonpath='{.data.ALPHA_VANTAGE_API_KEY}' | base64 --decode
```

Configuration Reference

Variable	Default	Description
`EXPERIMENT_MODE`	`OPTIMIZED`	`BASELINE` (plain LLM) or `OPTIMIZED` (RAG + agents)
`CONFIDENCE_THRESHOLD`	`0.70`	FinBERT minimum confidence to generate a report
`SIMILARITY_THRESHOLD`	`0.55`	ChromaDB cosine similarity minimum for retrieved docs
`ALPHA_VANTAGE_API_KEY`	—	Required for live market data

To switch to BASELINE mode, edit `banana-configmap.yaml`:

```yaml
data:
  EXPERIMENT_MODE: "BASELINE"
```

Then apply:

```bash
kubectl apply -f banana-configmap.yaml
kubectl rollout restart deployment/banana-api
```

Local Docker Alternative (No Kubernetes)

If you prefer running without Minikube:

```bash
export ALPHA_VANTAGE_API_KEY=your_key_here
docker-compose up --build
```

API will be available at `http://localhost:8000/analyze`.